

Possible Relationships

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Several of these variables are likely to reveal important relationships for long-term decarbonization goals and for forecasting national energy demand (note we are not limited to these if we discover better relationships):

Link Jakub's initial correlation matrix/data vs to justify choices

Josh

1. Wind Generation & Wind Speed

- Because wind power is expected to be a key contributor to net-zero efforts, how *wind_generation* and *wind_speed* vary alongside *national_demand* can highlight how much of the daily demand might be offset by wind resources and how reliably wind can meet peaks.
- Over time, you could see if rising wind capacity is lowering fossil-fuel-based demand or if there are seasonal/weather-driven fluctuations in wind production that need to be managed.

Adam

1. Solar Generation & Temperature/Rainfall

- *Solar_generation* directly measures solar's contribution to overall supply. Studying its relationship with *national_demand*, alongside *max_temp*, *min_temp*, and *rain_mm*, can reveal when solar is most effective (e.g. sunny days might show sharper midday drops in national demand if solar helps cover daytime load).
- If a large fraction of demand eventually shifts to being met by solar, you'd expect to see less correlation of demand with outside temperature or time-of-day peaks; or at least a shift in the net load shape.

Jakub (or Average Price Daily)

1. Temperature (min/max) and Rainfall

- Weather drives heating/cooling usage (especially temperature extremes), so *min_temp* and *max_temp* may correlate strongly with *national_demand*. A cold snap (low *min_temp*) raises heating load; a hot stretch can raise cooling load. Analyzing this in conjunction with *solar_generation*, *wind_generation*, and *average_price_daily* indicates how well renewables keep up under extreme weather.

Jacob (or Temperature)

1. Average Price Daily

- The *average_price_daily* variable connects supply-demand conditions with cost. If high renewable output (wind, solar) pushes prices down, then the shape of the *national_demand* curve could shift as consumers respond. This becomes highly relevant for long-run decarbonization—low prices during high renewable availability may spur demand shifting or storage adoption.

From a practical standpoint for forecasting, **temperature, wind speed, wind and solar generation, and price** are prime candidates for explaining fluctuations in **national demand**. And in the broader context of net-zero targets, how the system copes with renewable fluctuations (wind/solar) under different weather scenarios (temperature, wind speed, rainfall) will be central to understanding the balance between demand and green generation resources.